Eshara

Egyptian Sign Language Translator

Deliverables: Project Planning and Management, Literature Review, Requirements Gathering
Track: Microsoft Machine Learning Engineer

Team:

1. Merna Karam Abdullah Abdel Samie
2. Adham Hossam Abdel Moneim Abdel Badie
3. Salwa Wael Youssef Ali
4. Ahmed Mohamed Ibrahim Mohamed Abou El-Ezz
5. Fares Ehab Zaghloul

Supervisor:
Mostafa Asala

Part 1: Project Planning and Management

1.1 Project Proposal

Overview. Eshara is a translator for Egyptian Sign Language (ESL). A camera films the signer, deep learning models recognise the sign, and the app shows it as Arabic text and also reads it aloud. The aim is to make conversation easier between deaf and hearing people.

Problem. Many deaf and hard-of-hearing people use ESL, but most hearing people do not know it, so even simple conversations can be difficult. Most existing recognition systems are built for American Sign Language or for single letters, and they do not cover everyday Egyptian expressions.

Objectives

- Record our own ESL video dataset of everyday words and expressions, such as Hello, Thank You, Help, Eat, Drink, Tired, Yes, No, Good and Bad.
- Extract hand and pose landmarks from the videos with MediaPipe and OpenCV.
- Train LSTM and Transformer models on the landmarks and compare them.
- Turn the predicted sign into Arabic text and Arabic speech.
- Build a real-time web application with a backend API.
- Evaluate the models and report the results as they are, including the weak points.

Scope

In scope	Out of scope (for this phase)
Word and expression recognition from a single camera	Full sentence translation with grammar
Arabic text and Arabic speech output	Fingerspelling of the whole alphabet (possible future work)

In scope	Out of scope (for this phase)
Real-time web app (Streamlit) with a Flask or FastAPI backend	Native mobile app
Custom dataset with several signers, backgrounds and lighting conditions	Text or speech to sign (the reverse direction)

1.2 Project Plan

Official DEPI deadlines

Item	Deadline
Project Planning and Management	16 Oct 2026
Literature Review	16 Oct 2026
Requirements Gathering	16 Oct 2026
System Analysis and Design	6 Nov 2026
Implementation (source code and execution)	30 Nov 2026
Final presentation, testing and reports	4 Dec 2026

Milestones and deliverables

#	Milestone	Target date	Deliverable
M1	Planning, literature review and requirements done	16 Oct	This document, uploaded to GitHub
M2	System analysis and design done	6 Nov	Use case, ER, DFD, sequence, activity, state and class diagrams, wireframes, architecture
M3	Dataset collected and landmarks extracted	13 Nov	Dataset (kept outside the repo) and preprocessing scripts
M4	Model trained and evaluated	20 Nov	Trained LSTM and Transformer models, evaluation report
M5	Application integrated (API, web UI, TTS)	30 Nov	Source code, README, deployed or executable app
M6	Testing, reports and final presentation	4 Dec	Test plan, bug reports, user manual, technical documentation, PPT/PDF, infographic, optional demo video

Gantt chart (weekly, shaded cells show when the activity is active)

Activity	W1 12-16 Oct	W2 19-23 Oct	W3 26-30 Oct	W4 2-6 Nov	W5 9-13 Nov	W6 16-20 Nov	W7 23-27 Nov	W8 30 Nov-4 Dec
Planning, literature review, requirements								
System analysis and design								

Activity	W1 12-16 Oct	W2 19-23 Oct	W3 26-30 Oct	W4 2-6 Nov	W5 9-13 Nov	W6 16-20 Nov	W7 23-27 Nov	W8 30 Nov-4 Dec
Dataset collection and labelling								
Landmark extraction and preprocessing								
Model training and tuning								
Backend API and web app								
TTS and integration								
Testing and bug fixing								
Reports, presentation, infographic								

Resources

- Hardware: laptops and webcams for recording and the demo, plus GPU access (Google Colab or Kaggle) for training.
- Software: Python, OpenCV, MediaPipe, TensorFlow or PyTorch, Flask or FastAPI, Streamlit, Git and GitHub.
- People: the team, volunteers who can sign for the recordings, and if possible someone who knows ESL to check the sign labels.

1.3 Task Assignment and Roles

The roles below were agreed by the team. Everyone helps with data collection. Each member works on their own Git branch and opens pull requests so the others can review.

Role	Member	Responsibilities
Project Manager and Documentation	Merna Karam Abdullah Abdel Samie	Plan and deadlines, GitHub board, meetings, risk tracking, README and reports
Data Lead and QA	Salwa Wael Youssef Ali	Sign list, recording protocol, collection, labelling, dataset split, test plan, bug reports
Computer Vision Engineer	Adham Hossam Abdel Moneim Abdel Badie	OpenCV pipeline, MediaPipe landmark extraction, normalisation, augmentation
ML Engineer	Ahmed Mohamed Ibrahim Mohamed Abou El-Ezz	LSTM and Transformer models, training, tuning, evaluation
Backend and Frontend Engineer	Fares Ehab Zaghloul	Flask or FastAPI inference API, Streamlit web app, Arabic TTS, wireframes, accessibility

1.4 Risk Assessment and Mitigation Plan

#	Risk	Likelihood	Impact	Mitigation
R1	Not enough data or too few signers, so the model overfits	High	High	Fixed recording protocol, several signers and backgrounds, augmentation, split by signer
R2	Wrong sign labels, because we are not native ESL users	Medium	High	Ask ESL users or interpreters and check labels before training

#	Risk	Likelihood	Impact	Mitigation
R3	Poor accuracy with different lighting or camera positions	Medium	High	Record in varied conditions and use normalised landmarks instead of raw pixels
R4	Real-time latency too high on ordinary laptops	Medium	Medium	Small models, sliding-window inference, optimise the model if needed
R5	Similar signs get confused	Medium	Medium	Study the confusion matrix, add more samples, use both hands and pose features
R6	GPU or compute limits	Medium	Medium	Use Colab or Kaggle, keep models small, save checkpoints often
R7	Schedule slips because data collection takes longer	High	High	Start with a small core vocabulary and expand later, review progress weekly
R8	Privacy concerns about videos of people	Medium	Medium	Written consent, no raw videos on GitHub, keep landmarks only where possible
R9	Merge conflicts and lost work	Low	Medium	Feature branches, pull request reviews, regular commits
R10	A team member is unavailable	Low	Medium	Shared documentation, and every role has a backup

1.5 KPIs

These targets are provisional. We have not collected the dataset yet, so the accuracy targets will be confirmed after the first baseline training run.

KPI	Metric	Target
Recognition quality	Accuracy on a held-out test set	Initial aim of at least 80%, confirmed after the first baseline run
Recognition quality	Macro F1-score	Initial aim of at least 0.80, confirmed after the first baseline run
Generalisation	Accuracy on a signer not seen in training	Measured and reported, target set after the baseline
Responsiveness	Time from the end of a sign to the text	2 s or less
Real-time performance	Processing frame rate	At least 15 FPS on a standard laptop
Dataset	Number of classes, videos per class and signers	About 10 signs first; video and signer counts fixed after a pilot recording session
Reliability	Crash-free sessions in testing	At least 95%
Usability	User satisfaction (survey, 1 to 5)	At least 4.0
Adoption	Test users who finish a task successfully	At least 80%
Process	Milestones delivered on time	6 of 6

Part 2: Literature Review

2.1 Background

Sign languages are complete natural languages with their own grammar. They are expressed through hand shape, position, movement and orientation, and also through facial expression and body posture. Egyptian Sign Language is the language of Egypt's deaf community and it is different from other sign languages. Most published work and datasets are about ASL, German (RWTH-PHOENIX-Weather), Chinese and a few others. There is much less public data for Egyptian and Arabic sign languages, and that is the main reason we are building our own ESL dataset.

Work on sign languages is usually split into three tasks:

- Isolated sign recognition: classify one sign per video clip. This is what Eshara does.
- Continuous sign language recognition: recognise a sequence of signs in one video.
- Sign language translation: produce a spoken-language sentence, which has to deal with different grammar and word order.

2.2 Approaches in the Literature

Gloves and sensors. Data gloves and motion sensors give accurate hand data, but they are expensive and not practical for everyday use. A vision-based system only needs a camera.

CNNs on raw frames. Early deep learning work, for example Pigou et al. (2015), used convolutional networks on Kinect video to recognise 20 Italian gestures. These models can be accurate, but they are sensitive to the background, the lighting and the way the signer looks, and they need a lot of data.

Landmark and skeleton methods. Pose estimation tools extract keypoints, which removes the background and makes the models smaller and faster. MediaPipe (Lugaresi et al., 2019) and MediaPipe Hands (Zhang et al., 2020) track hands in real time on ordinary hardware, so they suit a laptop application. ST-GCN (Yan et al., 2018) treats skeleton sequences as graphs and has been used for action and sign recognition.

Sequence models. LSTM (Hochreiter and Schmidhuber, 1997) is the usual choice for modelling how landmarks change over time. Transformers (Vaswani et al., 2017) use self-attention to capture long-range dependencies and train in parallel. Camgoz et al. (2020) applied them to sign language recognition and translation.

Sign language translation. Camgoz et al. (2018) introduced end-to-end sign language translation on RWTH-PHOENIX-Weather 2014T. Later work uses Transformers and gloss-free methods, but these need large datasets that do not exist for ESL.

Egyptian Sign Language. There is some ESL research but not much. Elhagry et al. (2021) compared a CNN and a CNN-LSTM on 9 common ESL words. Abdelrazik et al. (2023) used MediaPipe landmarks with LSTM and GRU models, including a GRU with a modified activation function. Ismail et al. (2023) describe how to build a dataset for translating ESL into Arabic speech, and they say they did it because large ESL datasets are missing. A public Mendeley dataset (2025) has 55 ESL words with 10 videos each, recorded with consistent lighting and background. Together these papers show that landmarks with a sequence model work for ESL, and that the missing piece is more varied data.

2.3 Datasets

Dataset	Language	Type	Relevance
WLASL (Li et al., 2020)	American	Word-level videos, 2,000+ words, 100+ signers	Shows the scale needed for word-level recognition
RWTH-PHOENIX-Weather 2014T (Camgoz et al., 2018)	German	Continuous recognition and translation corpus	Standard benchmark for continuous recognition
ESL words (Elhagry et al., 2021)	Egyptian	9 common words	Small, but shows that ESL recognition is feasible
Mendeley "Egyptian Sign Language" (2025)	Egyptian	55 words, 10 videos per word, consistent lighting and background	Closest public resource, but little variation in signers and conditions
Arabic sign language datasets	Arabic (various)	Mostly alphabet images or small word sets	Useful as a reference, but usually not Egyptian and not everyday expressions
Eshara dataset	Egyptian	Word and expression videos from several signers	Our own contribution

2.4 Comparison of Techniques

Technique	Strengths	Weaknesses	Fit for Eshara
CNN on raw frames	No separate keypoint step	Sensitive to background, heavy, needs a lot of data	Low
MediaPipe landmarks + LSTM	Light, real-time, not affected by background	Depends on the tracker (occlusion, fast movement)	High (baseline)
MediaPipe landmarks + Transformer	Captures long-range context over time	Needs more data and more tuning	High (for comparison)
ST-GCN	Models the joint structure directly	More complex to implement	Medium (future work)
End-to-end translation models	Produce full sentences	Need large datasets	Low for this phase

2.5 Comparison with Existing Egyptian Sign Language Work

The table compares Eshara with the Egyptian Sign Language work we reviewed: what each one did, its main limitation, and how our project is different.

Work	What it did	Limitation	How Eshara differs
Elhagry et al. (2021)	Compared CNN and CNN-LSTM on 9 common ESL words	Very small vocabulary; a CNN on raw frames is sensitive to background and lighting	Larger planned vocabulary, landmark features, several signers and conditions
Abdelrazik et al. (2023)	MediaPipe landmarks with LSTM and GRU (including a GRU with a modified activation)	Compares recurrent models only	Same landmark idea, but we also compare a Transformer and test on an unseen signer

Work	What it did	Limitation	How Eshara differs
Ismail et al. (2023)	Dataset-building approach for ESL to Arabic speech	Focuses on building the dataset; no real-time web app described in the paper we reviewed	We add a real-time web app with a backend API, Arabic text and speech
Mendeley ESL dataset (2025)	55 ESL words, 10 videos per word	Consistent lighting and background; little variation in signers and conditions	Our dataset has several signers, backgrounds and lighting conditions, and is split by signer

2.6 Research Gap and Contribution

Most systems focus on ASL or on letters. ESL work exists but is still small: one study covers 9 words, and the public 55-word dataset has only 10 videos per word, all recorded in similar conditions. Eshara tries to fill this gap in four ways. We build our own ESL word and expression dataset. We use a light landmark pipeline that runs in real time on ordinary hardware. We compare LSTM and Transformer classifiers on the same data. And we deliver everything as a web application that gives Arabic text and speech.

2.7 Feedback and Evaluation (to be completed by the lecturer)

Area	Lecturer's assessment
Project idea and relevance	
Literature review quality	
Feasibility of scope and plan	
Other comments	

2.8 Suggested Improvements

- Keep the first vocabulary small and add classes step by step, so there is a working demo early.
- Include signers with different hand sizes, skin tones and clothing, and test on a signer the model has not seen.
- Add face landmarks, because facial expression carries grammar in sign languages.
- Add a confidence threshold and a "no sign" class so the app does not output wrong words.
- Later: fingerspelling for names, sentence-level grammar, and the reverse direction (Arabic to a sign avatar).
- Involve deaf users and ESL interpreters to validate the signs and give feedback.

2.9 Final Grading Criteria

The marks are set by the lecturer and DEPI.

Component	Marks (out of 100)
Idea and objectives	10
System analysis, requirements and design	15
Algorithms, methods and techniques	20
Implementation, coding and results	20

Component	Marks (out of 100)
Testing and documentation	10
Innovation and added value	10
Presentation and demo	10
Final submission and delivery	5
Total	100

2.10 References

1. Hochreiter, S., and Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735-1780.
2. Vaswani, A. et al. (2017). Attention Is All You Need. Advances in Neural Information Processing Systems (NeurIPS), 30.
3. Lugaresi, C. et al. (2019). MediaPipe: A Framework for Building Perception Pipelines. arXiv:1906.08172.
4. Zhang, F. et al. (2020). MediaPipe Hands: On-device Real-time Hand Tracking. CV4ARVR 2020 workshop; arXiv:2006.10214.
5. Pigou, L., Dieleman, S., Kindermans, P.-J., and Schrauwen, B. (2015). Sign Language Recognition Using Convolutional Neural Networks. Computer Vision, ECCV 2014 Workshops, pp. 572-578. DOI: 10.1007/978-3-319-16178-5_40.
6. Camgoz, N. C., Hadfield, S., Koller, O., Ney, H., and Bowden, R. (2018). Neural Sign Language Translation. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). DOI: 10.1109/CVPR.2018.00812.
7. Camgoz, N. C., Koller, O., Hadfield, S., and Bowden, R. (2020). Sign Language Transformers: Joint End-to-end Sign Language Recognition and Translation. CVPR. arXiv:2003.13830.
8. Li, D., Rodriguez Opazo, C., Yu, X., and Li, H. (2020). Word-level Deep Sign Language Recognition from Video: A New Large-scale Dataset and Methods Comparison. IEEE Winter Conference on Applications of Computer Vision (WACV), pp. 1448-1458. DOI: 10.1109/WACV45572.2020.9093512.
9. Yan, S., Xiong, Y., and Lin, D. (2018). Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition. AAAI Conference on Artificial Intelligence, 32(1). DOI: 10.1609/aaai.v32i1.12328.
10. Elhagry, A. et al. (2021). Egyptian Sign Language Recognition Using CNN and LSTM. arXiv:2107.13647.
11. Abdelrazik, M. A., Zekry, A., and Mohamed, W. A. (2023). Efficient Deep Learning Algorithm for Egyptian Sign Language Recognition. 33rd Conference of Open Innovations Association (FRUCT), pp. 3-8. IEEE.
12. Ismail, Y. et al. (2023). From Gestures to Audio: A Dataset Building Approach for Egyptian Sign Language Translation to Arabic Speech. IEEE conference paper.
13. Egyptian Sign Language (2025). Mendeley Data, v2. DOI: 10.17632/39tbt2jd7r.2.

Part 3: Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Role	Needs
Deaf and hard-of-hearing ESL users	Primary users	Accurate and fast translation of their signs into Arabic, and an app they can use without training
Hearing people (family, colleagues, service staff)	Primary users	Understand the signer through text and speech without knowing ESL
ESL interpreters and educators	Domain experts	Correct sign labels and respect for the language; they may also use it for teaching
Project team	Developers	Clear scope, usable tools, code that can be reproduced
Supervisor, lecturer and DEPI	Evaluators	Complete documentation, a working system and good engineering practice
Hospitals, banks, schools and government offices	Possible adopters	A reliable and simple tool for front-desk communication
Data contributors (signers)	Participants	Privacy, informed consent and proper use of their recordings

3.2 User Stories

1. As a deaf user, I want the app to recognise my sign in front of the camera, so that a hearing person can read what I said in Arabic.
2. As a hearing person, I want to hear the translation in Arabic, so that I do not have to keep looking at the screen.
3. As a user, I want to start and stop the camera easily, so that I control when I am being recorded.
4. As a user, I want to see a confidence level, so that I know when the result is uncertain.
5. As a user, I want to see a history of recent translations, so that I can go back over the conversation.
6. As a user, I want a simple, high-contrast Arabic interface, so that I can use it without help.
7. As a data contributor, I want to record and label signs with my consent, so that the dataset can grow.
8. As a developer or admin, I want to retrain or replace the model, so that accuracy can improve over time.
9. As a teacher, I want to see the list of supported signs, so that I can use the app for practice.

3.3 Use Cases

Actors: signer (deaf user), hearing user, data contributor, administrator or developer, camera, and the text-to-speech service.

ID	Use case	Primary actor	Summary
UC1	Translate a sign in real time	Signer	The camera captures the sign, landmarks are extracted, the model predicts the word and the Arabic text is shown
UC2	Hear the translation	Hearing user	The predicted Arabic text is spoken with text-to-speech
UC3	Control the camera	Signer or hearing user	Start, stop and choose the camera
UC4	View translation history	Any user	Look back at recently recognised words
UC5	View supported signs	Any user	Browse the vocabulary list
UC6	Record a training sample	Data contributor	Record a labelled sign video after giving consent
UC7	Train or update the model	Administrator	Run preprocessing and training, evaluate, and deploy a new model

UC1: Translate a sign in real time (detailed)

Precondition: the camera is available, permission is granted and the model is loaded.

Main flow: (1) The user starts the camera. (2) The system captures frames. (3) The system extracts hand and pose landmarks. (4) The system keeps a window of frames in a buffer. (5) The model predicts the sign and a confidence score. (6) If the confidence is above the threshold, the Arabic word is shown. (7) Optionally, the word is spoken.

Alternative flows: if no hands are detected, ask the user to move into the frame; if the confidence is low, show “not recognised” instead of a wrong word; if the camera fails, show a clear error message.

Postcondition: the recognised word is shown and added to the history.

3.4 Functional Requirements

ID	Requirement	Priority
FR1	The system shall capture live video from a camera.	Must
FR2	The system shall detect and extract hand and pose landmarks from each frame.	Must
FR3	The system shall classify a sequence of landmarks into one of the supported ESL words or expressions.	Must
FR4	The system shall display the recognised sign as Arabic text.	Must
FR5	The system shall convert the Arabic text to speech.	Must
FR6	The system shall show a confidence score and reject low-confidence predictions.	Should
FR7	The system shall let the user start, stop and pause recognition.	Must
FR8	The system shall keep a session history of recognised signs.	Should
FR9	The system shall list the supported signs.	Could
FR10	The system shall expose a backend API for prediction (Flask or FastAPI).	Must
FR11	The system shall provide tools to record, label and preprocess dataset videos.	Must

ID	Requirement	Priority
FR12	The system shall provide scripts to train and evaluate the models and report accuracy and confusion matrices.	Must
FR13	The system shall handle errors gracefully (no camera, no hands detected, model not loaded).	Must
FR14	The system shall allow the model to be replaced without changing the application code.	Should

3.5 Non-functional Requirements

Category	Requirement
Performance	Latency of 2 s or less after a sign is finished, and at least 15 FPS on a standard laptop
Accuracy	Target set after the first baseline run (see the KPIs)
Usability	Simple Arabic-friendly interface (RTL support), high contrast, large text, usable without training
Accessibility	Readable fonts and clear visual feedback for users who cannot hear the speech output
Reliability	The app should not crash when hands are missing or the camera is interrupted, and it should show clear error messages
Security and privacy	No raw video stored without consent; camera frames are processed locally or on a trusted server; no personal data in the public repository; HTTPS if deployed
Portability	Runs on Windows, macOS and Linux with Python and a standard webcam, and is accessible through a web browser
Maintainability	Modular code, coding standards, comments, README and tests, with version control on GitHub
Scalability	New signs can be added by collecting data and retraining, without redesigning the system
Compatibility	Works with common webcams and modern browsers

3.6 Assumptions and Constraints

- The vocabulary is limited to the everyday words and expressions collected in this project. The final class list will be set after data collection.
- The system recognises one signer at a time, facing the camera, with the hands visible.
- Training uses free or low-cost compute (Colab or Kaggle), and the demo runs on a standard laptop.
- The raw video dataset is not stored in the GitHub repository because of privacy and size.
- All documents are uploaded to GitHub, as the DEPI guidelines require.

Appendix A: Dataset Collection Plan

The team will build its own Egyptian Sign Language dataset. All numbers below are proposals and will be confirmed after a pilot recording session.

A.1 Initial Vocabulary

We start with a core vocabulary of 10 signs plus one “no sign” class, and add more if time allows. An ESL user or interpreter must check every sign before we start recording.

#	Sign	Arabic label
1	Hello	مرحبا
2	Thank You	شكرا
3	Help	مساعدة
4	Eat	أكل
5	Drink	شرب
6	Tired	تعبان
7	Yes	نعم
8	No	لا
9	Good	كويس
10	Bad	وحش
11	No sign (hands at rest or random movement)	لا توجد إشارة

A.2 Dataset Size

Item	Proposed value
Signers	At least 5, including at least one person fluent in ESL if possible
Repetitions per sign per signer	10 (5 in the standard setup, 5 in a varied setup)
Videos per sign	50
Total videos	About 550 (11 classes x 50)
Clip length	2 to 3 seconds

A.3 Recording Protocol

- Camera: laptop or phone webcam at 720p and 30 FPS, fixed at chest height, about 1 to 1.5 m from the signer.
- Framing: upper body, with the head and both hands always visible.
- Each clip starts and ends with the hands at rest, so the sign is clearly separated.
- Standard setup: plain background, good light, short sleeves or contrasting clothing.
- Varied setup: different backgrounds (plain wall, cluttered room), lighting (daylight, indoor lamp, dim room), distance and clothing.
- Include both right-handed and left-handed signers when possible.
- Record in the same order for every signer, check each clip right after recording, and re-record the ones that fail.

A.4 Naming and Folder Structure

```
dataset/
  raw/<sign_id>/S01_C1_R03.mp4   (signer 01, condition 1, repetition 03)
  landmarks/<sign_id>/S01_C1_R03.npy
  metadata.csv
```

The columns of metadata.csv are: video_id, sign_id, arabic_label, signer_id, condition, repetition, date, consent_signed, notes.

A.5 Data Split

We split the data by signer, so the test signer is never seen during training. With 5 signers, 3 are used for training, 1 for validation and 1 for testing. If time allows, we will also run leave-one-signer-out cross-validation to get a more reliable result.

A.6 Preprocessing

1. Extract hand and pose landmarks from every frame with MediaPipe.
2. Normalise the coordinates relative to the shoulders, so the signer's position and size matter less.
3. Resample every clip to a fixed number of frames (for example 30).
4. Save the landmarks as .npy files and keep the raw videos outside GitHub.
5. Augment the training data only (small shifts, scaling, noise, and mirroring where the sign allows it).

A.7 Quality Control

- Two team members review each new batch of clips for wrong signs, cut-off hands and bad lighting.
- An ESL user or interpreter checks the final sign list and a sample of the clips.
- We keep a log of removed or re-recorded clips in the notes column of metadata.csv.

A.8 Consent and Privacy

Every participant signs a consent form before recording. The form says that:

- The videos are used only for this graduation project and its evaluation.
- Raw videos are kept on a private team drive with limited access and are not uploaded to GitHub.
- The participant can ask for their videos to be removed at any time before the project ends.
- Participants under 18 need a parent or guardian signature.

A.9 Schedule

The Data Lead organises the sessions and keeps the metadata file. All team members take part in recording.

Week	Dates	Activity
W2	19-23 Oct	Confirm the sign list with an ESL user; pilot recording (2 signs, 2 signers) to test the protocol and the MediaPipe pipeline
W3-W4	26 Oct-6 Nov	Main recording sessions for all signs and signers
W5	9-13 Nov	Extra signers and conditions, quality review, final split and landmark extraction